# Multi-Scale Feature Fusion and Transformer Encoding for Enhanced Land Use Classification on EuroSAT

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***Abstract*—Land-use and land-cover (LULC) classification from multispectral satellite imagery remains challenging due to spectral variability, heterogeneous landscapes, and subtle inter-class similarities. This paper proposes a hybrid multi-scale learning framework that integrates deep convolutional feature extraction, attention-guided refinement, adaptive gated fusion, and Transformer-based contextual encoding for robust classification on the EuroSAT dataset. Multi-level features are extracted using DenseNet-121 and refined through channel and spatial attention blocks, followed by adaptive aggregation via a gated fusion module. A Transformer encoder models long-range spatial dependencies, while complementary features from multiple CNN backbones are optimized using Genetic Algorithm-based feature selection and classified with conventional machine-learning models. Experimental results show that the proposed Fusion + Transformer + GA framework outperforms individual backbone models, achieving a validation accuracy of 97.41% and a test accuracy of 97.36%, demonstrating its effectiveness for high-precision LULC classification on multispectral satellite imagery.**

***Index Terms*—Land-Use and Land-Cover (LULC) classification, multispectral satellite imagery, EuroSAT dataset, multi-scale feature fusion, attention mechanisms, Transformer encoder, genetic algorithm-based feature selection, remote sensing**

## I. Introduction

Land-use and land-cover (LULC) classification from satellite imagery plays a vital role in environmental monitoring, urban planning, agricultural assessment, and climate-related studies. With the increasing availability of multispectral remote-sensing data, benchmark datasets such as EuroSAT, derived from Sentinel-2 imagery, have become widely used for evaluating automated LULC classification methods. Despite the relatively high spatial and spectral resolution of such datasets, accurate classification remains challenging due to spectral variability, heterogeneous land-cover patterns, and subtle inter-class similarities across different categories.

Early LULC classification approaches primarily relied on traditional machine-learning algorithms, including support vector machines (SVM), k-nearest neighbors (KNN), decision trees (DT), and random forests (RF). While these methods demonstrated reasonable performance on simpler datasets, their dependence on handcrafted features significantly limits their ability to model complex spatial–spectral relationships present in real-world satellite imagery. As a result, their generalization capability deteriorates when confronted with large-scale and diverse land-cover distributions.

The advent of deep learning has substantially advanced LULC classification by enabling end-to-end feature learning directly from raw imagery. Convolutional neural network (CNN) architectures such as VGG, ResNet, DenseNet, and EfficientNet have shown strong performance by capturing hierarchical spatial features. More recent developments, including ConvNeXt architectures, hybrid CNN-based models, and data augmentation strategies leveraging generative adversarial networks (GANs), have further improved robustness and generalization. In addition, domain-specific enhancements such as vegetation and water indices (e.g., NDVI and NDWI) and global pooling strategies have contributed to improved land-cover discrimination.

Nevertheless, EuroSAT-based LULC classification continues to face key challenges related to class imbalance, intra-class spectral variability, and fine-grained inter-class similarities. Conventional CNNs often struggle to effectively integrate multi-scale information and long-range spatial context, which are essential for resolving ambiguities between visually similar land-cover classes. These limitations motivate the need for hybrid architectures that combine multi-scale feature representation, attention-driven refinement, and global contextual modeling.

To address these challenges, this paper proposes a hybrid

multi-scale learning framework that integrates deep convolutional feature extraction, attention-guided feature enhancement, adaptive gated fusion, and Transformer-based contextual encoding. By jointly leveraging complementary representations from multiple CNN backbones and optimizing feature selection using a genetic algorithm, the proposed approach aims to achieve robust and high-precision LULC classification on multispectral satellite imagery.

### A. Contributions of This Work

The key contributions of this work are:

- A hybrid multi-scale LULC classification framework that integrates convolutional feature hierarchies with Transformer-based global context modeling.
- Attention-enhanced multi-level feature extraction using channel and spatial attention blocks applied to intermediate DenseNet-121 representations.
- An adaptive gated fusion strategy with $1 \times 1$ projection for effective alignment and aggregation of multi-scale features.
- Transformer-based encoding of fused features to capture long-range spatial dependencies.
- Genetic algorithm–based feature selection for dimensionality reduction and improved classification performance.

## II. Literature Survey

The EuroSAT dataset, derived from Sentinel-2 multispectral imagery, has emerged as a standard benchmark for land-use and land-cover (LULC) classification in remote sensing [7]. Early studies primarily relied on classical machine-learning techniques such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), Decision Trees (DT), and Random Forests (RF). Although these methods achieved reasonable performance on low-complexity data, their dependence on handcrafted features limited their ability to capture complex spatial–spectral interactions, resulting in modest accuracies typically below 90% on EuroSAT [1].

#### *Deep Learning-Based EuroSAT Classification*

The adoption of deep learning significantly advanced EuroSAT classification by enabling end-to-end hierarchical feature learning. Convolutional neural networks (CNNs) such as VGG16, ResNet50, DenseNet, and EfficientNet demonstrated improved discriminative capability by learning multi-level spatial representations directly from imagery [8], [9], [10]. Karunasri *et al.* [2] reported accuracies up to 92.5% using ResNet50 and EfficientNet, highlighting the effectiveness of deep CNNs over classical approaches. However, these architectures remain constrained by fixed receptive fields, limiting their ability to capture long-range spatial context.

#### *Transformer-Based EuroSAT Models*

To overcome the locality bias of CNNs, recent studies explored transformer-inspired architectures for remote sensing. Models such as ConvNeXt incorporate large-kernel convolutions and normalization strategies to approximate global context modeling. While these approaches improve robustness, pure transformer-based solutions remain computationally intensive and often require large-scale training data, restricting their widespread adoption for multispectral EuroSAT imagery.

#### *Hybrid Architectures on EuroSAT*

Hybrid frameworks combining CNNs with adaptive or sequential models have shown promising results. Karunasri *et al.* [4] proposed a ConvNeXt–Liquid Neural Network (LNN) hybrid that achieved 97.26% accuracy by leveraging both convolutional feature extraction and dynamic temporal adaptability. These results indicate that hybridization effectively mitigates the limitations of standalone CNNs, though further improvements depend on more efficient feature fusion and contextual modeling strategies.

#### *Multi-Level Feature Learning and Domain Enhancements*

Several works emphasized enhancing CNN representations through pooling strategies, attention mechanisms, and domain-specific features. Uppalapati and Kaur [3] demonstrated that Global Sum Pooling (GSP) preserves richer activation information than traditional pooling, achieving 87.3% accuracy on EuroSAT. In parallel, spectral indices such as NDVI and NDWI were integrated with deep models to improve vegetation and water discrimination. Despite these gains, pooling and feature enhancement alone are insufficient to resolve fine-grained inter-class similarities.

#### *Generative and Other Related Approaches*

Generative models have been explored to address class imbalance and data scarcity. GAN-based augmentation techniques, including InfoGAN, ACGAN, and CDC-GAN, increased data diversity but yielded limited classification improvements, with accuracies typically below 85% [5]. More recent diffusion-based methods improved generative quality metrics but did not translate into superior classification performance [6]. Dataset distillation and self-supervised learning represent emerging directions, though their application to multispectral EuroSAT data remains limited.

#### *Summary and Research Gap*

Overall, existing EuroSAT literature demonstrates a progression from classical machine learning to deep CNNs, hybrid architectures, and generative augmentation techniques. While hybrid models currently achieve the strongest performance, challenges remain in effectively integrating multi-scale features, modeling long-range spatial dependencies, and reducing feature redundancy. These limitations motivate the proposed hybrid multi-scale framework that combines attention-enhanced feature extraction, adaptive fusion, transformer-based encoding, and genetic algorithm–based feature selection for high-precision LULC classification.

Table I presents a comprehensive comparison of existing EuroSAT-based LULC classification approaches, highlighting the evolution from classical models to hybrid deep learning frameworks and motivating the proposed methodology.

TABLE I
COMPREHENSIVE LITERATURE SURVEY OF LULC CLASSIFICATION APPROACHES ON THE EUROSAT DATASET.

| Category | Representative Methods | Core Idea | Dataset | Accuracy (%) |
|---|---|---|---|---|
| Classical Machine Learning | SVM, KNN, Decision Tree, Random Forest [1] | Handcrafted spectral–spatial features with shallow classifiers | EuroSAT | 81–86 |
| Deep CNN-Based Models | VGG16, ResNet50, DenseNet, EfficientNet [2] | Hierarchical convolutional feature learning | EuroSAT | 88–92.5 |
| Pooling and Feature Enhancements | CNN + Global Sum Pooling [3] | Preservation of activation strength in deep layers | EuroSAT | 87.3 |
| Hybrid Architectures | ConvNeXt + Liquid Neural Network [4] | Combination of convolutional hierarchy and adaptive dynamics | EuroSAT | 97.26 |
| Generative Augmentation | CDC-GAN, InfoGAN, ACGAN [5] | Synthetic data generation for class imbalance handling | EuroSAT | $\leq$85 |
| Diffusion-Based Models | Quantum Latent Diffusion Models [6] | Improved generative quality via latent diffusion | EuroSAT | – |
| **Proposed Method** | **Multi-Level Fusion + Transformer + GA** | **Attention-guided fusion, global context modeling, feature optimization** | **EuroSAT** | **97.36** |

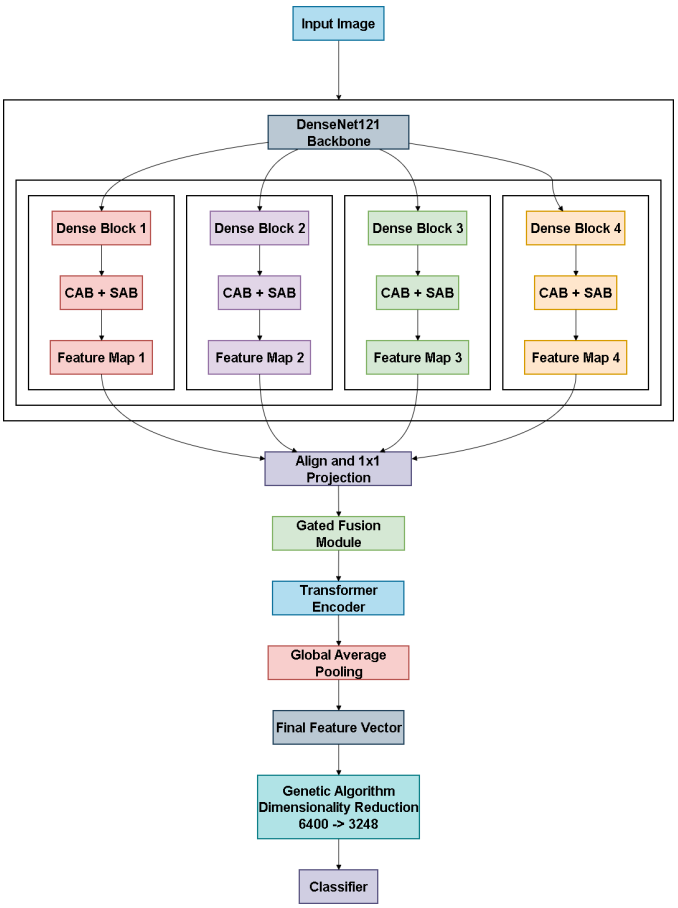


Fig. 1. Overall architecture of the proposed Multi-Level DenseNet Fusion + Attention + Transformer + GA Framework.

## III. METHODOLOGY

The proposed framework follows a hybrid learning pipeline that integrates multi-level convolutional feature extraction, attention-guided fusion, Transformer-based contextual modeling, and genetic algorithm–based feature selection for robust LULC classification on the EuroSAT dataset [7]. An overview of the proposed architecture, illustrating the multi-level feature extraction, attention-guided fusion, Transformer encoding, and genetic algorithm–based feature selection, is shown in Fig. 1.

### A. Data Preparation and Feature Extraction

Given an input multispectral image $x \in \mathbb{R}^{H\times W\times C}$, images are resized and normalized using ImageNet statistics. DenseNet-121 is employed as the primary backbone due to its dense connectivity and feature reuse capability [9]. Multi-scale feature maps are extracted from Dense Blocks 1–4:

$$F_i = \phi_i(x), \quad i \in \{1, 2, 3, 4\}.$$

In addition, pretrained CNNs including ResNet50, VGG16, EfficientNet-B0, MobileNetV2, and AlexNet are used as frozen feature extractors to obtain complementary global representations:

$$f_k = \psi_k(x), \quad k = 1, \ldots, K.$$

### B. Attention-Guided Multi-Level Fusion

Each multi-level feature map $F_i$ is refined using channel and spatial attention mechanisms to enhance discriminative responses [11], [12]. To enable fusion, all refined features are spatially aligned and projected to a unified channel dimension using $1 \times 1$ convolutions:

$$\tilde{F}_i = W_i * F_i'.$$

An adaptive gated fusion module learns spatial importance weights $\alpha_i$ using a softmax function:

$$\alpha_i = \frac{\exp(g_i)}{\sum_{j=1}^{4} \exp(g_j)},$$

and produces the fused representation:

$$F_{\text{fused}} = \sum_{i=1}^{4} \alpha_i \tilde{F}_i.$$

### C. Transformer Encoding and Feature Integration

The fused feature map is reshaped into a sequence of tokens and augmented with positional encodings. A Transformer encoder is employed to model long-range spatial dependencies [13]:

$$H = \text{Transformer}(F_{\text{fused}}).$$

Global average pooling is applied to obtain a compact embedding:

$$v = \frac{1}{N} \sum_{n=1}^{N} H_n.$$

This embedding is concatenated with features from the additional CNN backbones to form a unified representation:

$$X = [f_1, f_2, \ldots, f_K, v].$$

### D. Genetic Algorithm-Based Feature Selection and Classification

Due to the high dimensionality of $X$, a genetic algorithm is employed to identify an optimal feature subset [14]. Each chromosome is encoded as a binary mask $m \in \{0,1\}^{|X|}$, and feature selection is performed as:

$$X_m = X \odot m.$$

The fitness of each chromosome is evaluated using classification accuracy:

$$\text{Fitness}(m) = \text{Acc}(h(X_m)),$$

where $h(\cdot)$ denotes a lightweight classifier. The selected features are finally classified using Support Vector Machines, Logistic Regression, and Random Forests.

## IV. Summary

The proposed methodology unifies multi-level DenseNet feature learning, attention-based fusion, Transformer-driven contextual modeling, and genetic algorithm–based feature optimization into a compact hybrid framework, achieving robust and high-precision LULC classification on the EuroSAT dataset.

## V. Experimental Results and Analysis

This section presents a comprehensive evaluation of the proposed hybrid framework on the EuroSAT dataset. The objective of these experiments is to assess the effectiveness of multi-level feature fusion, Transformer-based contextual modeling, and genetic algorithm–based feature selection in comparison with individual CNN backbone models. All experiments are conducted using a 70/15/15 train–validation–test split to ensure fair and consistent performance assessment across models.

### A. Performance Comparison

Table II summarizes the validation and test accuracies obtained by different CNN backbone models paired with their respective classifiers, along with the proposed hybrid approach. Among the individual backbones, DenseNet-121 combined with Logistic Regression achieves the highest test accuracy of 96.77%, which can be attributed to dense feature reuse and improved gradient propagation. EfficientNet-B0 and ResNet-50 also demonstrate strong performance, whereas lightweight architectures such as MobileNet-V2 and AlexNet yield comparatively lower accuracies due to limited representational capacity.

The proposed Fusion + Transformer + GA framework consistently outperforms all individual backbone models, achieving a validation accuracy of 97.41% and a test accuracy of 97.36%. This performance gain highlights the advantage of integrating multi-scale convolutional features with attention-guided fusion and global contextual modeling. The inclusion of genetic algorithm–based feature selection further contributes to robustness by eliminating redundant features while preserving discriminative information.

### B. Classification Performance Analysis

The proposed model achieves an average F1-score of 0.97 across all land-use categories, indicating a balanced trade-off between precision and recall. Minor misclassifications are primarily observed between visually and spectrally similar classes, such as *AnnualCrop* and *HerbaceousVegetation*, which is consistent with the intrinsic ambiguity of the EuroSAT dataset. Nevertheless, the model maintains stable performance across all categories, demonstrating strong generalization capability.

### C. Genetic Algorithm Convergence and Feature Reduction

The convergence behavior of the genetic algorithm indicates that an optimal feature subset is identified within ten generations. The selected subset reduces the feature dimensionality by approximately 49%, from 6,400 to 3,248 features, without any degradation in classification accuracy. This result confirms the presence of substantial redundancy in multi-backbone feature representations and demonstrates the effectiveness of evolutionary optimization for compact feature selection.

## VI. Discussion

The experimental results confirm that the proposed hybrid framework provides a consistent performance advantage over individual CNN backbones and existing hybrid architectures. Multi-level feature extraction from DenseNet-121, enhanced by channel and spatial attention mechanisms, improves discriminative power by emphasizing relevant spectral and spatial cues. The adaptive gated fusion module further strengthens representation learning by selectively integrating information across scales, while the Transformer encoder captures long-range spatial dependencies that are difficult to model using convolution alone.

TABLE II
PERFORMANCE COMPARISON ON THE EUROSAT DATASET USING A 70/15/15 TRAIN–VALIDATION–TEST SPLIT.

| CNN Model | Classifier | Val. Acc. (%) | Test Acc. (%) |
|---|---|---|---|
| ResNet-50 | Logistic Regression | 96.17 | 95.90 |
| DenseNet-121 | Logistic Regression | 96.89 | 96.77 |
| EfficientNet-B0 | Logistic Regression | 95.51 | 95.73 |
| MobileNet-V2 | SVM | 93.29 | 93.88 |
| AlexNet | SVM | 91.88 | 91.95 |
| **Fusion + Transformer + GA + LR** | | **97.41** | **97.36** |

The genetic algorithm–based feature selection significantly improves computational efficiency by reducing feature dimensionality without compromising accuracy. Even after dimensionality reduction, the proposed method surpasses strong hybrid baselines such as ConvNeXt–LNN, indicating superior feature integration and generalization. Remaining classification errors are largely confined to classes with highly similar spectral signatures, suggesting that future extensions incorporating full multispectral bands or additional domain-specific indices may further enhance performance. Overall, these findings validate the effectiveness of combining multi-level attention refinement, adaptive fusion, Transformer-based contextual encoding, and evolutionary optimization for high-precision land-use and land-cover classification on the EuroSAT benchmark.

## VII. CONCLUSION AND FUTURE WORK

This paper presented a hybrid learning framework for land-use and land-cover classification that integrates multi-level DenseNet feature extraction, attention-guided refinement, adaptive gated fusion, Transformer-based global context modeling, and genetic algorithm–based feature selection. By jointly leveraging multi-scale spatial–spectral representations and long-range contextual dependencies, the proposed approach achieves a test accuracy of 97.36% on the EuroSAT dataset, outperforming strong CNN and hybrid baselines. The evolutionary feature selection strategy further reduces feature dimensionality by nearly 50% without sacrificing classification performance, demonstrating both robustness and computational efficiency.

Future work will explore extending the framework to fully exploit the multispectral bands of Sentinel-2 data and investigating end-to-end fine-tuning of Transformer components for enhanced contextual learning. Additional directions include evaluating the proposed approach on larger and more diverse remote-sensing benchmarks such as RESISC45 and AID, as well as incorporating spectral indices and advanced data augmentation techniques to further improve discrimination among visually similar land-cover classes.

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